

Ahmed Tarek Mourssi

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SUMMARY

STEAM & Computer Science Educator with 10 years of experience delivering hands-on, inquiry-based learning in robotics, physical computing, and programming. Bachelor's degree in Computer Science with expertise in project-based curriculum design across diverse international settings including UAE, USA, Saudi Arabia, and Egypt. Proficient in Arduino, Python, Scratch, and microcontroller integration. Achieved 1st place at RoboCup Junior Egypt 2017 and led students to international robotics competitions. Specialized in EdTech integration, makerspace learning, and fostering inclusive STEM environments for learners of all ages.

I have Dubai Freelancing Residence + valid driving license.

EXPERIENCE

STEAM Session Lead

Udacity | DECI - Digital Egypt Cubs Initiative with The Egyptian Government

August 2022 - January 2026, USA

- Delivered 100+ virtual STEAM lessons in Computer Fundamentals, AI, and Python programming to students across various levels for 3000+ Students
- Designed and implemented hands-on coding and electronics projects, improving student engagement and practical learning outcomes by 30%.
- Conducted weekly assessments with structured feedback, leading to a 20% improvement in student performance.

STEAM & Robotics Instructor

Alpha M Educational and Sporting Services Ltd

July 2024 - September 2024, Dubai, UAE

- Taught robotics and programming to teenagers in top-tier schools such as American School of Dubai (ASD), JESS, and Kings' Schools.
- Developed and facilitated interactive, project-based learning modules, increasing student engagement by 40%.
- Introduced students to robotics engineering concepts, fostering a strong foundation in STEM education.

STEAM & ICT Teacher

Innova Schools

- Delivered STEAM and ICT curriculum online, integrating inquiry-based learning with hands-on coding, electronics, and robotics projects to foster student engagement and critical thinking.
- Designed and facilitated project-based learning modules using Python, Scratch, Arduino, physical computing, and sensors to develop computational thinking and problem-solving skills.
- Integrated EdTech tools and LMS platforms (Google Classroom, Zoom, Microsoft Teams) to create engaging virtual learning environments and ensure 100% accessibility to digital resources.
- Taught hands-on coding, electronics, and robotics through virtual labs and simulation tools, adapting makerspace activities for online delivery.
- Collaborated with interdisciplinary teams to integrate STEAM principles across curricula, promoting holistic learning experiences.

AWASIS Science

ICT Instructor (Hybrid)

July 2016 - July 2020, Alexandria, Egypt

- Instructed ICT and science curricula using inquiry-based learning, integrating Python, robotics, electronics, and AI into hands-on STEAM experiences.
- Facilitated makerspace activities involving microcontrollers, sensors, and automation, improving student digital literacy and technology proficiency.
- Taught in hybrid environments, adapting lessons for classroom, remote, and blended learning settings across diverse student groups.
- Managed EdTech tools including LMS platforms (Google Classroom, Moodle, Blackboard), ensuring 100% accessibility to digital resources.
- Achieved 1st place in Best Designs at Robotics Competitions and represented the team at an international competition in Nagoya, Japan.

Super Tutor | Computer Science Tutor

Preply

February 2020 - March 2022

- Delivered 1,000+ hours of tutoring in Python, AI, physical computing, and robotics to students globally.
- Designed personalized, hands-on lesson plans with Arduino, Scratch, and electronics projects, resulting in a 95% student satisfaction rate.
- Facilitated inquiry-based learning experiences integrating coding with hardware components and sensors.

EdTech Software Engineer & STEAM Instructor (Top 3% Worldwide)

Upwork

September 2020 - October 2024

- Managed AI-integrated edtech projects, leading development teams to deliver scalable learning solutions.
- Achieved \$80K+ in freelance earnings by providing customized e-learning platforms and automation tools.
- Top 3% Worldwide Wide

Lego Robotics – MIT app Inventor Instructor

Robokids

February 2022 – July 2022, Alexandria Egypt

- Combined visual programming platforms with robotics kits to provide interactive coding experiences aligned with STEAM education standards.
- Demonstrated robotics workflows and the integration of sensors and microcontrollers, allowing students to practice collaborative decision-making and competition strategy formulation for WRO events.
- Applied LEGO Mindstorms and microcontroller platforms in guided workshops, instructing students in robot assembly, sensor deployment, and system automation to elevate participation and engagement.
- Led students to compete in WRO, where they applied engineering principles and physical computing to solve real-world challenges.

Python Instructor

almentor

July 2021 – March 2022, Cairo, Egypt

- Taught Python programming and computational thinking to 650+ students on Almentor.net, one of the largest e-learning platforms in the Middle East.
- Developed a structured STEAM curriculum integrating coding with problem-solving projects, leading to a 25% increase in student retention and completion rates.
- Created hands-on programming exercises that fostered inquiry-based learning and practical application of computer science concepts.

Python Developer – SaaS

modernint(Linker Product)

January 2023 – December 2023, Saudi Arabia · Remote

- Developed Linker Product using Python to automatically apply for LinkedIn job postings, resulting in a 40% increase in job application efficiency.

Junior Python Developer

Peakregroup

August 2022 – January 2023, United States · Remote

- Designed a product for scraping real estate data from USA states using Python, cleaning and formatting the data for input into Google Sheets, resulting in a more efficient and accurate data management process.
- Applied programming and automation skills to solve real-world data challenges, demonstrating technical versatility.

Contestant | 1st Place Robocup Egypt & Participant at Nagoya Japan International Competition

RoboCup Federation

July 2017 – August 2017, Nagoya, Aichi, Japan

- Designed and programmed autonomous robots that successfully completed complex challenges, securing 1st place in the RoboCup Egypt competition by optimizing algorithms for superior decision-making and task execution.

EDUCATION

Bachelor's degree, Computer Science

El Shorouk Academy · July 2023

CS50 Introduction to AI

HarvardX

Introduction to Machine learning

UDACITY · 2020-2023

CS50 Introduction to Computer Science

HarvardX · Online · 2020

CERTIFICATIONS

Level 2 Safeguarding Children Course

emtrg.co.uk · 2024

SKILLS

STEAM Education & Pedagogy: Project-Based Learning, Inquiry-Based Learning, Hands-On Learning Experiences, STEM/STEAM Curriculum Development, Makerspace & Innovation Environments, Workshop & Camp Facilitation, Interdisciplinary Projects, Educational Technology Integration, Differentiated Instruction, Student Assessment & Progress Tracking

Programming & Physical Computing: Python, Scratch, C++, C, Java, JavaScript, Arduino, Raspberry Pi, ESP32, micro:bit, Microcontrollers & Sensors, Electronics & Circuits, Prototyping, PCB Design (Eagle PCB), Embedded Systems, IoT Development, Robotics Programming, Computer Vision, OpenCV, Machine Learning

Tools & EdTech Platforms: Google Classroom, Moodle, Blackboard, Django REST Framework, SOLIDWORKS, CAD Software, 3D Printing Technologies, Robotics Kits (LEGO Mindstorms, VEX), ROS (Robot Operating System), Selenium, SQL, MySQL, REST APIs